1 Models

1.1 Arrivals

Abstract `Arrival` interface defines stochastic envelope for arrivals:

$$P(A(s,t) > \alpha(t-s,\sigma)) < \varepsilon(\sigma)$$

Each arrival class must implement the following functions:

- $\alpha(t,\sigma)$: `curve_func(self)` $\rightarrow$ `Callable[..., float]`;
- $\varepsilon(\sigma)$: `err_func(self)` $\rightarrow$ `Callable[..., float]`.

1.1.1 Token bucket arrivals

Class `TokenBucketArrival` has rate r and burst b as parameters, and implements the following functions:

- $\alpha(t,\sigma) = r \cdot t + b + \sigma$;
- $\varepsilon(\sigma) = 0$.

1.1.2 Discrete Markov-modulated On-Off arrivals

Describes a MMOO EBB traffic. Starts at its steady state distribution.

Class `DiscreteMMOOArrival` has $stay_on$, $stay_off$ and $peak_rate$ as parameters, and implements the following functions:

- $\alpha(t,\sigma,\theta) = r(\theta) \cdot t + b(\theta) + \sigma$, where:
  - $r(\theta) = \frac{1}{\theta} \log \left(\frac{p_{00}+p_{11}+\sqrt{(p_{00}+p_{11}e^{\theta P})^2-4(p_{00}+p_{11}-1)e^{\theta P}}}{2} \right)$ with $stay_on = p_{11}$, $stay_off = p_{00}$ and $peak_rate = P$;
  - $b(\theta) = \frac{1}{\theta} \log \left(e^{\theta P} \cdot \frac{\max\limits_{k \in \overline{1,M}} \{v_k\}}{\min\limits_{k \in \overline{1,M}} \{v_k\}} \cdot \frac{2}{p_{00}+p_{11}+\sqrt{(p_{00}+p_{11}e^{\theta P})^2-4(p_{00}+p_{11}-1)e^{\theta P}}} \right)$ with v_k - k-th component of the eigenvector $\boldsymbol{v}$ of matrix $\varphi(\theta)T$ (diagonal matrix containing MGFs of the traffic in each state multiplied by transition probability matrix;
- $\varepsilon(\sigma,\theta) = e^{-\theta\sigma}$.

1.2 Services

Abstract `Service` interface defines a stochastic envelope for a service:

$$P(D(t) < A * [\beta - \sigma]^+(t)) < \varepsilon(\sigma)$$

Each service class must implement the following functions:

- `curve_func(self)` $\rightarrow$ `Callable[..., float]`;
- `err_func(self)` $\rightarrow$ `Callable[..., float]`.

1.2.1 Constant-rate service

Class `ConstantRateService` has rate R and burst b (default: 0) as parameters, and implements the following functions:

- $\beta'(t) = [Rt + b - \sigma]^+$;
- $\varepsilon(\theta) = 0$